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About Me

I am a PhD student in Intelligent Robotics at Sungkyunkwan University advised by Prof. Hyeonwoo Yu, where I am a member of the Lab of Artificial Intelligence and Robotics (Lair). My research centers on 3D spatial perception for robotics, and I am currently interested in language-grounded and physics-aware representations of 3D space.

Education

MS/Ph.D	Sungkyunkwan University , Intelligent Robotics • Advisor: Prof. Hyeonwoo Yu	Mar. 2023 – Present
B.S.	UNIST , Computer Science and Mechanical Engineering	Mar. 2016 – Feb. 2023

Experience

Naver Labs , Research Intern • Vision Group research 3D Vision & Deep Learning internship. • Published work at RA-L 2025. • Supervisor: Giseop Kim, Sunwook Choi	Aug. 2024 – Feb. 2025
University of Illinois Urbana-Champaign , Global Program Participant • Global Program Participant in AI & Big Data at UIUC.	Jan. 2022 – Feb. 2022
UNIST , Applied Cryptography Lab Research Intern • Research internship in Applied Cryptography Lab. • Supervisor: Miran Kim	June 2021 – July 2022

Publications

TriLift: Interpolation-Free Tri-Plane Lifting for Efficient 3D Perception on Embedded Systems • Sibaek Lee , Jiung Yeon, Hyeonwoo Yu • <i>In Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i>	2026
LangGS-SLAM: Real-Time Language-Feature Gaussian Splatting SLAM • Seongbo Ha, Sibaek Lee , Kyeongsu Kang, Joonyeol Choi, Seungjun Tak, Hyeonwoo Yu • <i>arXiv Preprint</i>	2026
LEGO-SLAM: Language-Embedded Gaussian Optimization SLAM • Sibaek Lee , Seongbo Ha, Kyeongsu Kang, Joonyeol Choi, Seungjun Tak, Hyeonwoo Yu • <i>arXiv Preprint</i>	2025
LAMP: Implicit Language Map for Robot Navigation • Sibaek Lee , Hyeonwoo Yu, Giseop Kim, Sunwook Choi • <i>IEEE Robotics and Automation Letters (RA-L)</i>	2025
Spatial Coordinate Transformation for 3D Neural Implicit Mapping • Kyeongsu Kang, Seongbo Ha, Sibaek Lee , Hyeonwoo Yu • <i>IEEE Robotics and Automation Letters (RA-L)</i>	2025
Bayesian NeRF: Quantifying Uncertainty With Volume Density for Neural Implicit Fields • Sibaek Lee , Kyeongsu Kang, Seongbo Ha, Hyeonwoo Yu • <i>IEEE Robotics and Automation Letters (RA-L)</i>	2025

Just flip: Flipped observation generation and optimization for neural radiance fields to cover un-observed view2024

- **Sibaek Lee**, Kyeongsu Kang, Hyeonwoo Yu
- *In Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*

Projects	
Artificial Intelligence Innovation Graduate School (Sungkyunkwan University)	2026 – 2031
• supported by IITP funded by the Korea government (MSIT)	
Training of Experts in AI Product Application Technologies for Advanced Industries	2026 – 2031
• supported by KIAT funded by the Ministry of Trade, Industry and Energy	
Physical AI-Based Integrated Autonomous Navigation for AMRs	2026
• industry-academia collaboration project funded by Samsung Electronics GTR	
Development of a Multimodal Humanoid Foundation AI Model for Real-World Manipulation and Locomotion	2025 – 2027
• supported by IITP grant funded by the Korea government (MSIT)	
Expansion of the Humanoid AX project applicable to manufacturing sites and daily life	2025 – 2028
• supported by KEIT funded by the Ministry of Trade, Industry and Energy	
AI Star Fellowship Support Program	2025 – 2029
• Supported by IITP grant funded by the Korea government (MSIT)	
AI Semiconductor Innovation Center	2025 – 2030
• Supported by IITP grant funded by the Korea government (MSIT)	
Development of Robotic Technology for Grasping and Manipulating Challenging Thin Non-Rigid Objects in Unstructured Environments Multi-Modal 3D Perception & Task Planning Learned from Demonstrations and Manuals for Battery Pack Recycling	2024 – 2027
• Funded by the Ministry of Trade, Industry and Energy	
Development and Demonstration of Unmanned Autonomous Operation Technology Based on Field-Use Visualization Sensors and 6-Axis Rotational Angle Sensors	2024 – 2028
• Funded by the Ministry of Trade, Industry and Energy	
Spatial and Task Generalization Framework for Hardware Platform-Agnostic Embodied AI	2024 – 2027
• Funded by the National Research Foundation of Korea	

Awards	
Best Paper Award (LG AI Research prize)	Aug. 2025
<i>Korean Artificial Intelligence Association (CKAIA)</i>	

Languages	
Korean (Native)	English (Professional)